Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukh

ABSTRACT

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks. We propose the Transformer, a novel network architecture based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and

1. Introduction & Background

Recurrent neural networks, long short-term memory (LSTM), and gated recurrent neural networks have been firmly established as state-of-the-art approaches in sequence modeling. Recurrent models inherently factor computation along the symbol positions of input and output sequences. This sequentially generates a sequence of hidden states h_t, preventing parallelization within training examples, which becomes critical at longer sequence lengths.

Attention mechanisms have become an integral part of compelling sequence modeling, allowing modeling of dependencies without regard to their distance in the input or output sequences. In this work we propose the Transformer, an architecture eschewing recurrence and instead relying entirely on self-attention to compute representations of input and output.

2. Model Architecture: Scaled Dot-Product & Multi-Head Attention

The Transformer follows an encoder-decoder structure using stacked self-attention and point-wise, fully connected layers for both encoder and decoder. The encoder is composed of a stack of N = 6 identical layers. Each layer has two sub-layers: a multi-head self-attention mechanism, and a simple position-wise fully connected feed-forward network.

Scaled Dot-Product Attention: We compute attention on queries Q, keys K, and values V as: Attention(Q, K, V) = softmax(Q * K^T / sqrt(d_k)) * V. The scaling factor 1/sqrt(d_k) counteracts extremely small gradients for large dimensions d_k.

Multi-Head Attention allows the model to jointly attend to information from different representation subspaces at different positions: MultiHead(Q,K,V) = Concat(head_1, ..., head_h) * W^O, where head_i = Attention(Q * W_i^Q, K * W_i^K, V * W_i^V). In our work we employ h = 8 parallel attention heads.

3. Empirical Results on WMT 2014 & Computational Complexity

On the WMT 2014 English-to-German translation task, the big transformer model achieves a state-of-the-art BLEU score of 28.4, outperforming the best existing models including ensembles by more than 2.0 BLEU. Training took 3.5 days on 8 NVIDIA P100 GPUs, a fraction of previous training costs.

On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days, at 1/4 the training cost of previously published models.

Limitations & Complexity: Self-attention per layer has computational complexity O(n^2 * d) where n is sequence length and d is representation dimension. When sequence length n exceeds dimension d, self-attention layers are slower than recurrent layers, necessitating future work for extreme context lengths.